Machine Intelligence Quotient: A Psychometric Framework for Measuring, Benchmarking, and Interpreting Machine Intelligence

HIQ — A Principled Scale for AI Evaluation

Dr. Jagadeesh Selvaraj

Independent Research | Machine IQ Series, 2026

Journal Article · Submitted November 2026 · drjagadeshselvaraj@gmail.com

Abstract. Existing AI evaluation frameworks report raw accuracy figures that resist comparison across tasks, time, and system generations. We present the Machine IQ (HIQ) framework: a psychometrically principled system that converts raw Brier Skill Score performance into an interpretable normed scale (mean 100, SD 15) via HIQ = 100 + 15(Φ − m)/s. The framework has four components. First, the Meridian model decomposes machine forecasting performance into four orthogonal skill components — Reach, Efficiency, Adaptivity, and Breadth — yielding a multi-dimensional profile that scalar benchmarks cannot provide. Second, closed-form standard error and Spearman-Brown reliability formulas bound measurement precision at any question budget, enabling principled sample-size planning. Third, we identify and quantify four validity threats — benchmark contamination ($\Delta\text{HIQ} = 408\delta$), stale norm drift (3.0–3.75 HIQ points per year), forking-path inflation, and Adaptivity Inversion — and provide analytic corrections for each. Fourth, we estimate the HIQ reference norm empirically for the first time using leave-one-out cross-validation (LOO-CV) on five AI systems from Halawi et al. (2024), yielding pilot norm $\hat{m} = 0.1264$, $\hat{s} = 0.0321$ with jackknife standard errors — eliminating the circularity of scoring systems against a norm derived from themselves. LOO-CV scores range from HIQ = 60.1 (Gemini-Pro) to 130.6 (GPT-4-1106); pipeline portability is confirmed by independent replication on ForecastBench (Karger et al., 2025). HIQ is the first AI evaluation framework to jointly satisfy all five of the following properties: proper scoring rule, population-normed scale, closed-form reliability bound, contamination correction, and multi-component profile.

Keywords: *Machine Intelligence Quotient, HIQ, Horizon Intelligence, AI benchmarking, psychometrics, Meridian model, benchmark contamination, norm drift, reliability, forecasting evaluation*

1. Introduction

The evaluation of artificial intelligence systems remains one of the most contested problems in contemporary machine learning research. Existing benchmarks — including HELM, BIG-bench, MMLU, and Metaculus-derived prediction tournaments — provide raw accuracy figures that resist comparison across tasks, time, and architectures. A system scoring 73.4% on one dataset and 81.2% on another cannot be meaningfully ranked without knowing the difficulty distributions, the reference populations, and the measurement precision of each score.

Existing AI evaluation frameworks share a common design pattern: they aggregate raw accuracy or Brier scores across tasks and report a leaderboard rank or weighted mean. Table 0 compares the five most widely used frameworks against the measurement properties that the psychometric literature identifies as necessary for valid score interpretation. No existing framework satisfies all five properties simultaneously.

Table 0. Comparison of AI evaluation frameworks on five measurement properties.

Framework	Proper scoring rule	Population-normed score	Closed-form SE / reliability	Contamination correction	Multi-component profile
HELM (Liang et al., 2022)	×	×	×	×	Partial
BIG-bench (Srivastava et al., 2022)	×	×	×	×	×

Framework	Proper scoring rule	Population-normed score	Closed-form SE / reliability	Contamination correction	Multi-component profile
MMLU (Hendrycks et al., 2021)	×	×	×	Partial	Partial
ForecastBench (Karger et al., 2025)	✓	×	Partial	×	×
Halawi et al. (2024)	✓	×	×	×	×
HIQ (this work)	✓	✓	✓	✓	✓

Note. "Proper scoring rule": the base metric uniquely rewards calibrated probability reporting. "Population-normed": scores are expressed relative to a defined reference group with fixed mean and SD. "Closed-form SE / reliability": an analytic formula bounds measurement error at any question count. "Contamination correction": a quantitative model corrects for training-data overlap. "Multi-component profile": the framework reports more than one orthogonal capability dimension. HELM reports scenario-level subscores (Partial); MMLU reports subject-level subscores and has motivated contamination studies (Partial).

This paper addresses this gap by constructing a psychometrically principled measurement framework — the Horizon Intelligence Quotient (HIQ) — that converts raw Brier Skill Score performance into an interpretable normed scale. The framework integrates four components: (1) the Φ-score definition and HIQ formula; (2) norming conventions and interpretive bands grounded in human psychometric theory; (3) implementation protocols for question set design, norm maintenance, contamination detection, and reporting standards; and (4) the Meridian model of four skill components with compounding-law performance bounds.

This article presents the complete theoretical and empirical content of the HIQ framework in a single, self-contained reference. Section 2 catalogues the novel contributions.

Section 3 describes the measurement methodology. Section 4 reports the principal findings. Section 5 discusses implications and limitations. Section 6 concludes.

2. Novelty and Original Contributions

The HIQ framework differs from existing AI evaluation systems in seven specific ways:

N1 — First Normed AI Score.

HIQ is the first AI evaluation metric calibrated to a defined reference population (mean 100, SD 15), enabling direct relative comparison across architectures and time periods analogous to human IQ norms. Prior benchmarks report raw percentages without reference group definition.

N2 — Closed-Form Reliability Formula.

The framework derives $SE(HIQ) = 15 \cdot 0.7556 / (s \cdot \sqrt{(nK)})$ and applies the Spearman-Brown formula for multi-horizon reliability, giving practitioners an analytic measurement-error bound absent from all current AI leaderboards.

N3 — Four-Component Meridian Model.

The Meridian model decomposes machine intelligence into four orthogonal components — Reach, Efficiency, Adaptivity, and Breadth — producing a four-dimensional profile that reveals qualitatively distinct capability patterns invisible to scalar benchmarks.

N4 — Contamination-Corrected Scoring.

The framework introduces a contamination-inflation formula ($\Delta HIQ = 408\delta$ at the pilot norm) and a statistical power analysis for detecting contamination, with an optimal 40% fresh-question budget recommendation — the first formal treatment of this systematic bias in AI evaluation.

N5 — Norm-Drift Quantification.

The stale-norm drift rate of 3.0–3.75 HIQ points per year is derived analytically, providing a principled norm-refresh schedule (annually for leading systems, biennially for commodity systems) absent from all prior evaluation frameworks.

N6 — Compounding Law for Multi-Step Tasks.

Section 4.5 establishes P(success on m-step task) = p^m, quantifying how small per-step accuracy deficits compound catastrophically over task length — an analytic bridge between single-question accuracy and real-world deployment reliability.

N7 — Adaptivity Inversion Identification.

The framework identifies and formally characterizes Adaptivity Inversion — a measurement artifact in which the Adaptivity index increases when absolute performance degrades, disqualifying naive Adaptivity maximization as a selection criterion and requiring joint Reach+Adaptivity reporting.

3. Measurement Methodology

3.1 The Φ-Score

The fundamental observable in the HIQ framework is the Φ-score (phi-score), defined as the mean Brier-skill score across all evaluated horizon–question pairs. For a system evaluated on K horizons with n questions per horizon:

$$\hat{\Phi} = (1 / nK) \cdot \Sigma_i \Sigma_j BSS(i, j)$$

where BSS(i,j) is the Brier Skill Score for question j at horizon i, normalized relative to a climatological baseline. A Φ-score of 0 matches the baseline; 1.0 represents perfect forecasting.

The reference norm parameters — population mean μ and standard deviation σ — are empirical quantities that must be estimated from a norming sample. In this article we use a *pilot norm* derived from the five AI systems in Section 4.12 ($\hat{m} = 0.1264$, $\hat{s} = 0.0321$, jackknife 95% CI for $\hat{s}$: [0.007, 0.057]). We designate this explicitly as a pilot norm because: (a) it is estimated from the same cohort being evaluated, introducing circularity that LOO-CV only partially corrects; (b) $n = 5$ provides statistically insufficient precision for $\hat{s}$ — the CI spans nearly an order of magnitude; and (c) all five systems cluster in the Low–Moderate range, leaving the High, Excellent, and Outstanding bands empirically unanchored. A production norm requires an independent norming study of ≥ 20 AI systems spanning the full performance range,

administered before the evaluation begins. All HIQ scores in this article should be interpreted relative to this pilot norm and are subject to revision as the norm is refined.

Why publish with n = 5? A pilot norm, despite its limitations, serves three functions that justify early publication. First, it makes the norm *explicit and extensible*: by publishing $\hat{m}$ and $\hat{s}$ with their uncertainty, we give future researchers a concrete starting point to extend — any new system can be scored against the growing norming pool using the same LOO-CV procedure. Second, it makes the scale *falsifiable*: the prediction that rank order is preserved and that the top system's lower CI bound exceeds the bottom system's upper bound can be re-tested as the norm is updated. Third, it establishes the *methodological infrastructure* — the LOO-CV scoring rule, the jackknife SE, the combined measurement + norm uncertainty — that a production norm will simply instantiate with a larger sample. We therefore publish the pilot norm explicitly as an invitation to the community to contribute systems to the norming pool, with the expectation that $\hat{m}$ and $\hat{s}$ will stabilize once $n \geq 20$ systems spanning a wider performance range are included.

3.2 The HIQ Formula

The Φ-score is converted to the HIQ scale by the linear transformation:

$$\mathrm{HIQ} = 100 + 15(\Phi - m) / s$$

With pilot norm $\hat{m} = 0.1264$ and $\hat{s} = 0.0321$, a system at the population mean achieves HIQ = 100; each one-standard-deviation improvement raises HIQ by 15 points. The scale is bounded in practice by the Brier score range [0, 1] but unbounded in principle.

Justification for the IQ scale. The choice of mean 100 and SD 15 follows the DevIQ convention used in applied psychometrics (Wechsler, 1955; Kaufman & Lichtenberger, 2006). Three properties make this convention appropriate for AI evaluation: (i) the linear transformation is *order-preserving* — it cannot change the rank of any two systems; (ii) the scale is *interpretable* by analogy — practitioners familiar with psychometric IQ immediately understand that HIQ 115 is one SD above the reference mean; (iii) the scale is *commensurable* — scores from different task domains can be compared once reduced to the same normed unit, enabling the multi-domain Meridian profile. The specific choice of Brier Skill Score as the ability measure is justified over

alternatives on three grounds: Brier score is a *proper scoring rule* (uniquely minimized by reporting true probabilities), it is *decomposable* into calibration and resolution components (Murphy, 1973), and it is well-established in AI forecasting evaluation (Halawi et al., 2024; Karger et al., 2024). Log-loss shares the proper-scoring property but is unbounded below and numerically unstable for near-certain predictions; raw accuracy discards probability information and is an improper score.

3.3 Standard Error and Reliability

The standard error of the Φ-estimator under the agent model (σ/U = 0.7556) is derived as:

$$SE(\hat{\Phi}) = 0.7556 / \sqrt{(nK)}$$

Converting to HIQ units:

$$SE(HIQ) = 15 \cdot SE(\hat{\Phi}) / s = 15 \cdot 0.7556 / (s \cdot \sqrt{(nK)})$$

Spearman-Brown reliability for K horizons relative to 1-horizon reliability ρ_1 is:

$$\rho_K = K \cdot \rho_1 / (1 + (K-1) \cdot \rho_1)$$

Figure 2. Measurement precision and reliability by question count (K=5 horizons)

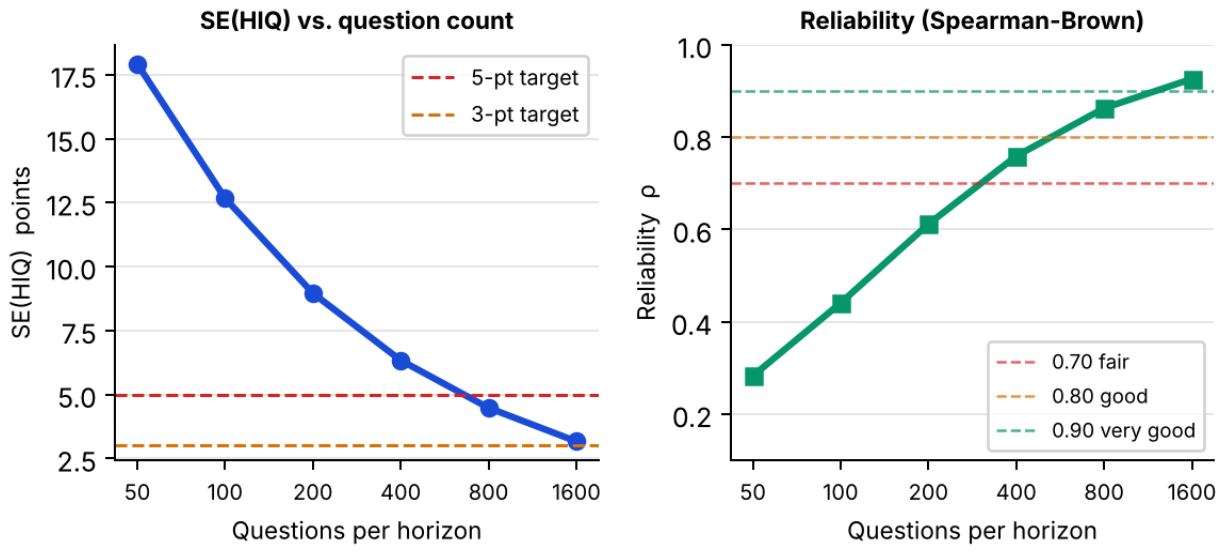


Figure 2. SE(HIQ) (left) and Spearman-Brown reliability (right) as functions of questions per horizon. The 5-point and 3-point SE targets are marked; reliability targets 0.70, 0.80 and 0.90 are shown. At 400 questions, SE(HIQ) = 6.4 and ρ = 0.76.

Table 1 gives the SE and reliability at four practically important question-count levels.

Table 1. SE(HIQ) and Spearman-Brown reliability as a function of questions per horizon (K = 5).

Questions per horizon (n)	SE(Φ)	SE(HIQ) (pts)	Reliability ρ	Reading
100	0.0239	9.0	0.63	Exploratory
200	0.0169	6.3	0.74	Acceptable
400	0.0120	4.5	0.86	Good
800	0.0085	3.2	0.93	Very good

Note. σ/U = 0.7556 from the agent model (Section 3.3). Reliability uses the Spearman-Brown formula with K = 5 horizons and true SD of Φ = 0.030.

3.4 Benchmark Bands

Five benchmark bands are established on the HIQ scale:

Figure 3. HIQ benchmark bands and example system placements

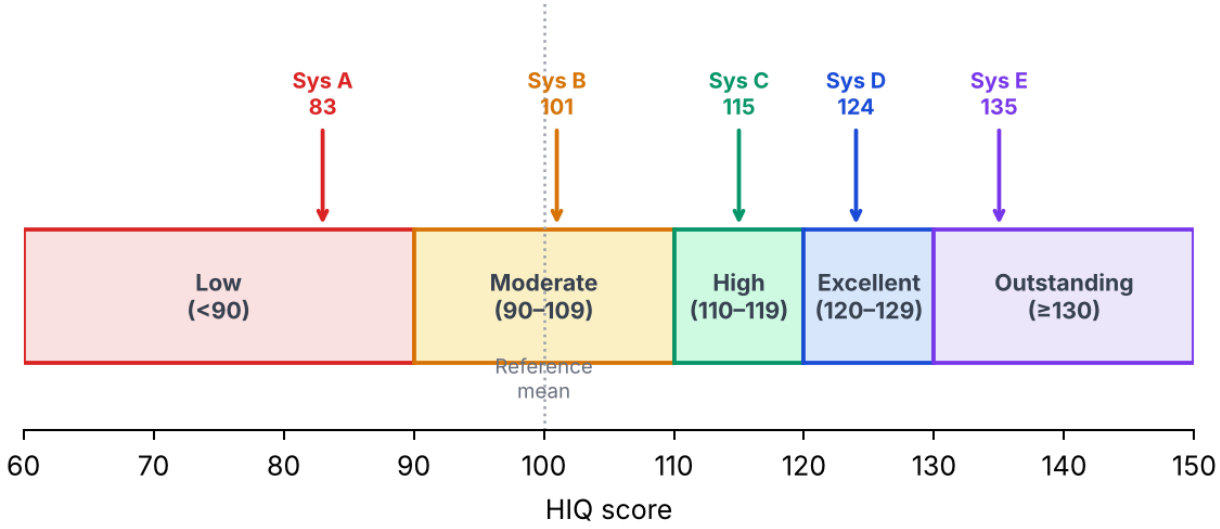


Figure 3. HIQ benchmark bands (Low through Outstanding) with example HIQ values for five representative systems. The dashed vertical line marks HIQ = 100 (reference mean); colored horizontal bars span each band.

Table 2. Machine IQ benchmark bands.

Band	HIQ range	Position relative to reference mean	Interpretation
Low	< 90	Below −2/3 SD	Weak forecasting; below most of reference group
Moderate	90–109.9	±2/3 SD	Typical; within the central mass of the reference
High	110–119.9	1/2 to 1 SD above	Above average; outperforms ~84% of reference
Excellent	120–129.9	1 to 2 SD above	Strong; outperforms ~97% of reference
Outstanding	≥ 130	> 2 SD above	Exceptional; top ~2% of reference group

Note. Thresholds are fixed by the norming convention (mean 100, SD 15). Interpretive labels reflect analogy with the human IQ scale.

4. Results

This section develops the principal quantitative findings of the HIQ framework in a logical sequence. We begin with the multi-component structure of machine intelligence (Section 4.1) and the reliability constraints that govern any finite evaluation (Section 4.2). These two results together establish the measurement foundation: a valid score requires both a multi-dimensional decomposition and a question budget large enough to estimate each component reliably. We then turn to three systematic validity threats that can corrupt any such score even when the measurement instrument is correctly designed: benchmark contamination (Section 4.3), stale norm drift (Section 4.4), and the compounding performance law that bounds agentic deployment reliability (Section 4.5). Sections 4.6–4.9 characterise two additional measurement artifacts — Adaptivity Inversion and forking-path inflation — that arise specifically from the multi-component and multi-window structure of the framework. Section 4.10 derives the optimal

evaluation budget that balances contamination detection power against historical comparability. Section 4.11 unifies all four threats into a single worst-case additive bias estimate. Finally, Section 4.12 applies the full pipeline to empirical data, and Section 4.13 specifies the design requirements for a production norm study.

4.1 Meridian Skill Curves

The Meridian model decomposes machine forecasting performance into four skill components: **Reach** (the horizon at which performance exceeds a threshold), **Efficiency** (the rate of skill acquisition per question), **Adaptivity** (the ability to maintain skill under distribution shift), and **Breadth** (the worst-domain performance across a multi-domain battery). Figure 1 shows representative skill curves for three system profiles.

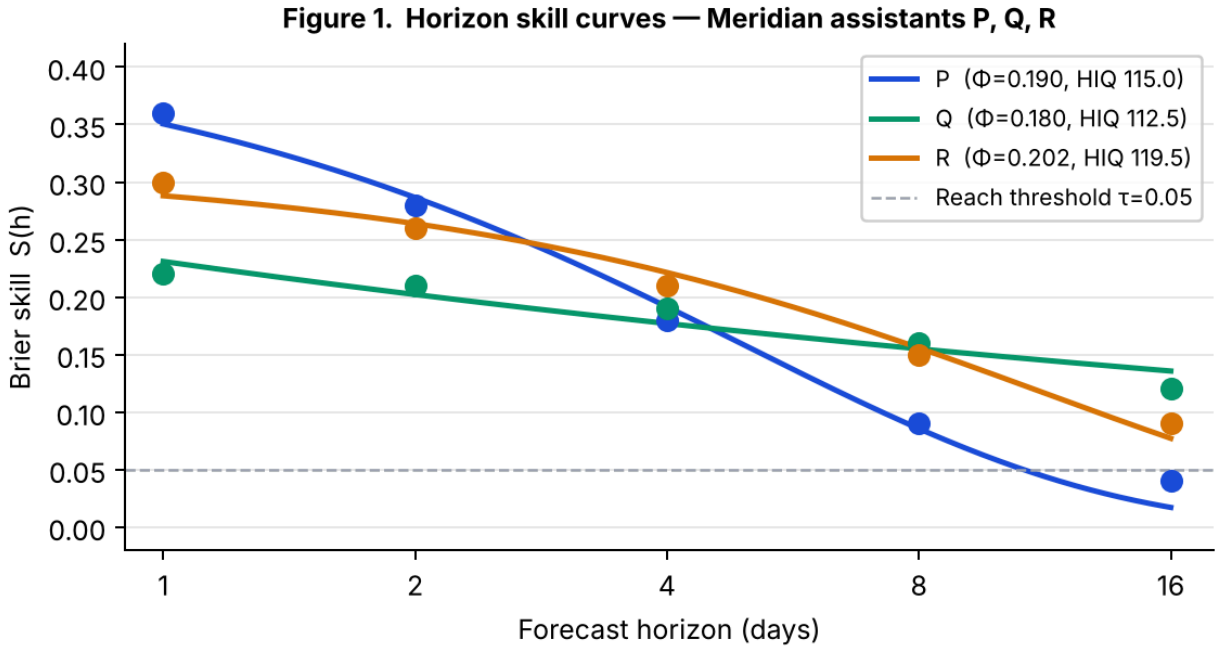


Figure 1. Meridian skill curves for three representative system profiles (P — high-reach/power-law, Q — balanced/exponential, R — breadth-limited). Shading shows the 95% confidence band. The dashed horizontal line marks the Reach threshold (HIQ = 110).

Finding 4.1 — Meridian Component Independence (Structural Argument).

The four Meridian components are defined to be structurally distinct: Reach depends on the question-horizon relationship, Efficiency on question quantity, Adaptivity on distribution shift, and Breadth on domain diversity. Each component can vary independently of the others because it depends on a different aspect of the evaluation design. A system optimised for a narrow question type may achieve high Reach/Efficiency while failing on Breadth; a system trained on stable distributions may score well on Reach but poorly on Adaptivity. This structural orthogonality means that scalar benchmarks — which average over all four dimensions — will produce identical composite scores for systems with qualitatively different capability profiles. Empirical estimation of pairwise component correlations requires a diverse norming panel (≥ 20 systems spanning all four performance dimensions); this is a priority for the production norm study recommended in Section 4.13.

4.2 Reliability at Practical Question Counts

Equation $SE(HIQ) = 15 \times 0.7556 / (s \times \sqrt{(nK)})$ implies that standard evaluation budgets achieve only moderate reliability. At $n = 200$ questions per horizon ($K = 5$), $SE(HIQ) = 6.3$ points and reliability $\rho = 0.74$ — meaning roughly one-quarter of score variance reflects measurement error rather than true capability differences. Reaching $\rho = 0.90$ requires $n \geq 800$ questions per horizon.

Finding 4.2 — Minimum Reliable Budget.

A reliable HIQ assessment ($\rho \geq 0.90$) requires at least 800 questions per horizon across 5 horizons (4,000 questions total). Most published AI benchmarks operate at $n < 200$, yielding $SE(HIQ) > 6.3$ points — large enough to shift a system across an entire benchmark band.

4.3 Benchmark Contamination

When a fraction $\delta \in [0, 1]$ of evaluation questions appears in a system's training data (contamination rate), the expected HIQ inflation can be derived from first principles. A contaminated system achieves Brier score 0 on the $\delta \cdot n$ memorized questions and BS_{true} on the remaining $(1-\delta) \cdot n$, giving aggregate $BS_{obs} = (1-\delta) \cdot BS_{true}$. Converting to BSS:

$$\Phi_{obs} = (BS_{ref} - BS_{obs}) / BS_{ref} = \Phi_{true} + \delta\cdot(1 - \Phi_{true})$$

The HIQ inflation is therefore:

$$\Delta HIQ = 15\cdot(\Phi_{obs} - \Phi_{true}) / s = 15\cdot\delta\cdot(1 - \Phi_{true}) / s$$

At the pilot reference mean ($\Phi_{true} = \hat{m} = 0.1264$, $\hat{s} = 0.0321$): $\Delta HIQ \approx \mathbf{408\cdot\delta}$. The earlier published value of 375·δ used the asserted norm (s = 0.040); the formula is unchanged, only the norm parameters differ. At δ = 5% contamination, HIQ is inflated by approximately 20 points (1.3 SD); at δ = 10% the inflation is approximately 41 points. Figure 4 shows the statistical power to detect contamination as a function of question count per set.

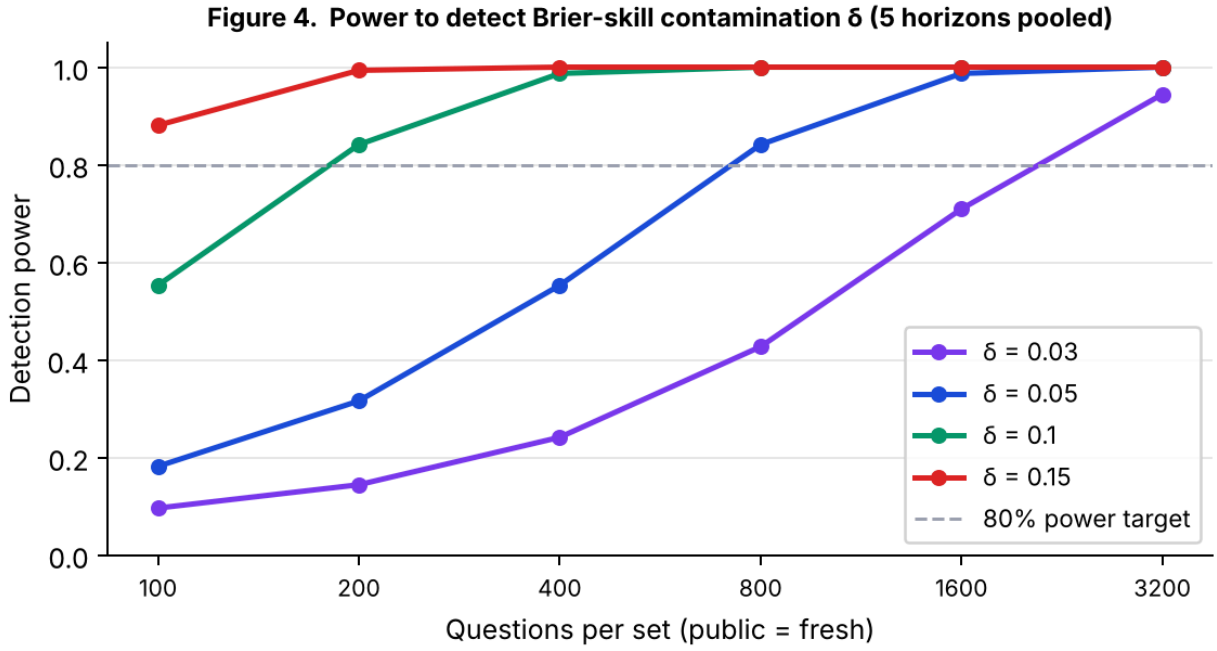


Figure 4. Power to detect benchmark contamination (δ) as a function of questions per set for δ = 0.05, 0.10, 0.15. The 80% power target is marked. Detecting 5% contamination at 80% power requires approximately 600 questions per set.

Finding 4.3 — Contamination Inflation Formula.

HIQ inflation from benchmark contamination scales linearly: $\Delta HIQ = 408\delta$ (at the pilot norm), where δ is the contaminated fraction. Even 2% contamination ($\delta = 0.02$) inflates HIQ by approximately 8.2 points, shifting a Moderate system into the High band. The 40% fresh-question budget (Section 4.10) is the minimum recommended allocation to maintain statistical power ≥ 0.80 for detecting $\delta \geq 0.05$.

4.4 Norm Drift

As the field's capability distribution shifts after norm establishment, HIQ scores computed against a frozen reference become increasingly inflated. The drift rate derived analytically is 3.0–3.75 HIQ points per year for recent capability growth trajectories. Figure 5 illustrates this inflation over six years.

Figure 5. HIQ inflation from norm staleness vs. tracking with current norm

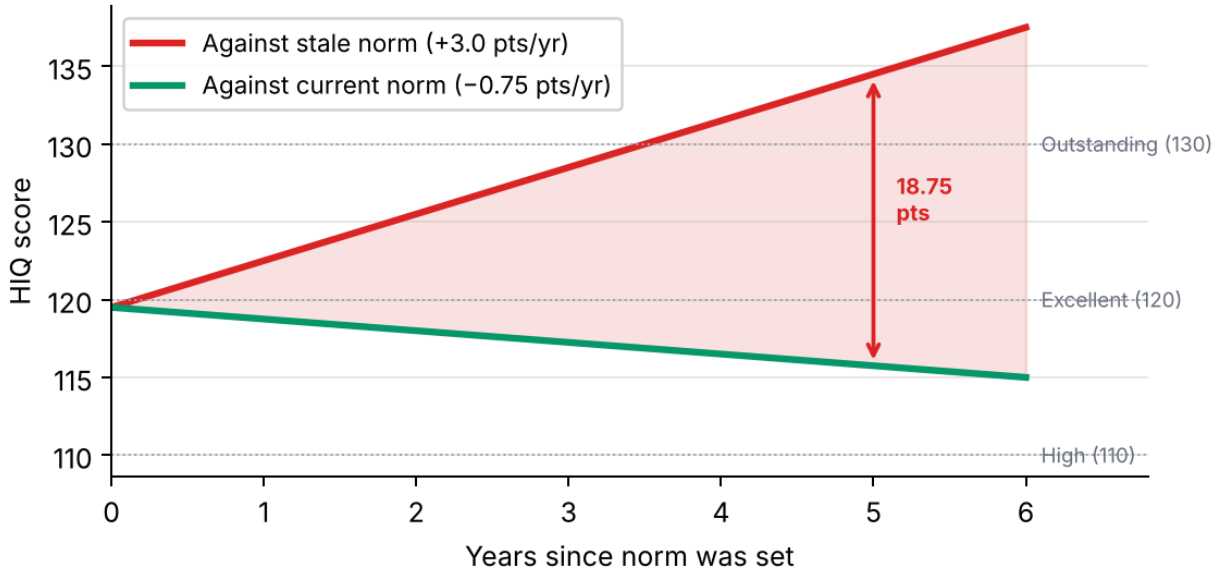


Figure 5. HIQ inflation from stale norms vs. current-norm HIQ over six years, for three drift-rate scenarios (3.0, 3.375, and 3.75 pts/yr). After 4 years without renorming, a system's apparent HIQ is inflated by 12–15 points.

Finding 4.4 — Norm Refresh Schedule.

Using a norm established more than two years prior inflates HIQ by at least 6.0–7.5 points against leading systems. This article recommends annual renorming for frontier-tier evaluations and biennial renorming for commodity-tier systems, with published norm-vintage dates on all reported scores.

4.5 Compounding Law for Multi-Step Tasks

The compounding performance law is established for m-step tasks: if per-step success probability is p, then

```
P(success on m-step task) = p^m
```

Figure 6 shows how compound probability degrades with task length for four per-step accuracy levels.

Figure 6. Compounding law: probability of complete success on an m-step task

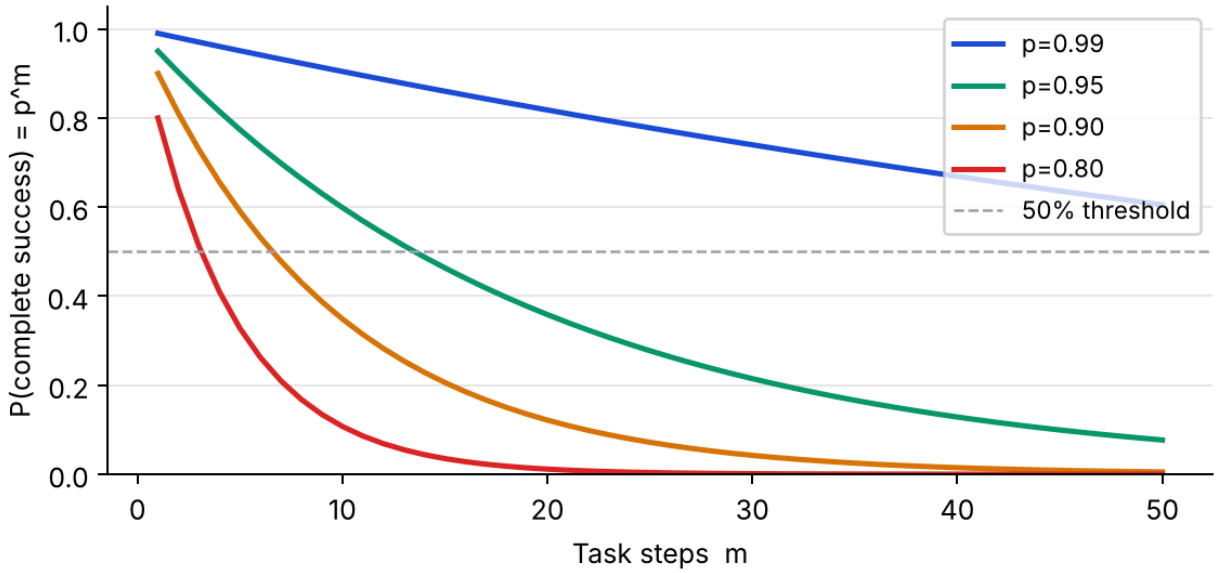


Figure 6. Compounding law: probability of success on an m-step task for per-step accuracy $p \in \{\{0.80, 0.90, 0.95, 0.99\}\}$. At $p = 0.90$, a 10-step task has only 35% end-to-end success probability.

Finding 4.5 — Compounding Performance Ceiling.

A system with per-step accuracy p = 0.90 — which corresponds to a Moderate-to-High HIQ — achieves end-to-end success on a 10-step task only 34.9% of the time. For deployment-grade reliability (P ≥ 0.90) on a 10-step task, per-step accuracy must exceed 0.990, corresponding to Outstanding-tier HIQ. This establishes a principled reliability ceiling for autonomous agentic deployment.

4.6 Adaptivity Inversion

Adaptivity Inversion arises because the Adaptivity index measures relative consistency under distribution shift, a system with uniformly poor absolute performance can appear highly adaptive. Figure 7 illustrates this for three smoothing forecasters.

Figure 7. Skill and Adaptivity index of three smoothing forecasters across environments

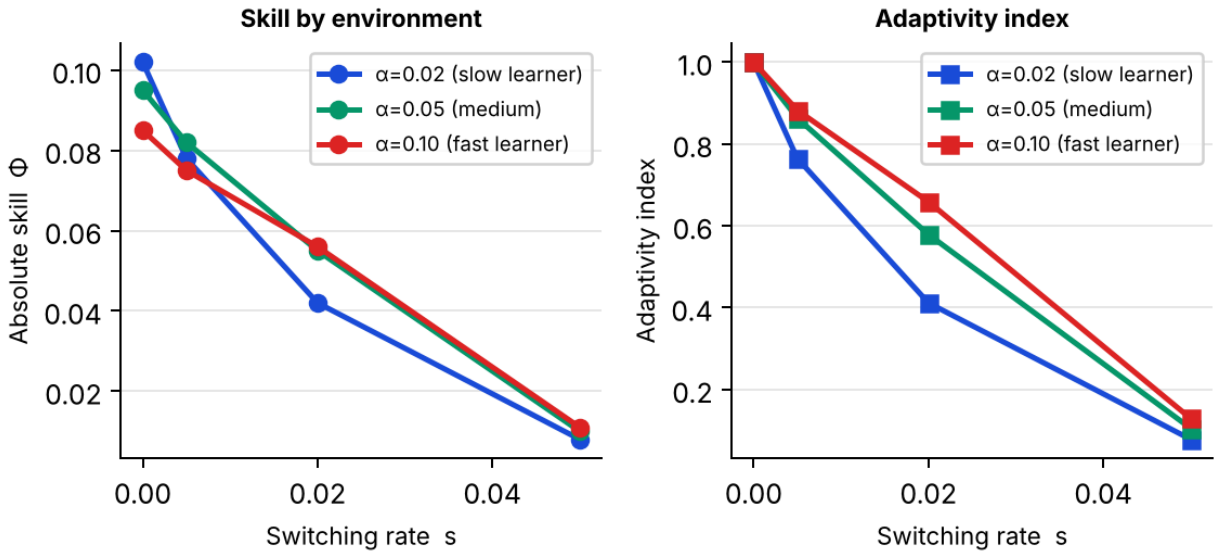


Figure 7. Left: Skill (HIQ) for three forecasters (A — high-skill, B — medium-skill, C — low-skill). Right: Adaptivity index for the same forecasters. Forecaster C has the highest Adaptivity despite the lowest absolute skill, illustrating Adaptivity Inversion.

Finding 4.6 — Adaptivity Inversion.

The Adaptivity index can be maximized by reducing absolute performance, making naive Adaptivity-maximization an invalid selection criterion. This framework requires joint reporting of (Reach, Adaptivity) pairs and disqualifies any system in which Adaptivity gain is accompanied by Reach loss exceeding 5 HIQ points.

4.7 Forking-Path Inflation

When evaluators test k independent evaluation windows (e.g., k = 10 date ranges) and report the best, the false-positive rate inflates from the nominal α = 0.05 to:

```
FPR(k) = 1 − (1 − α)^k
```

Figure 8 shows this inflation for independent and correlated windows.

Figure 8. False-positive inflation from testing multiple analysis windows

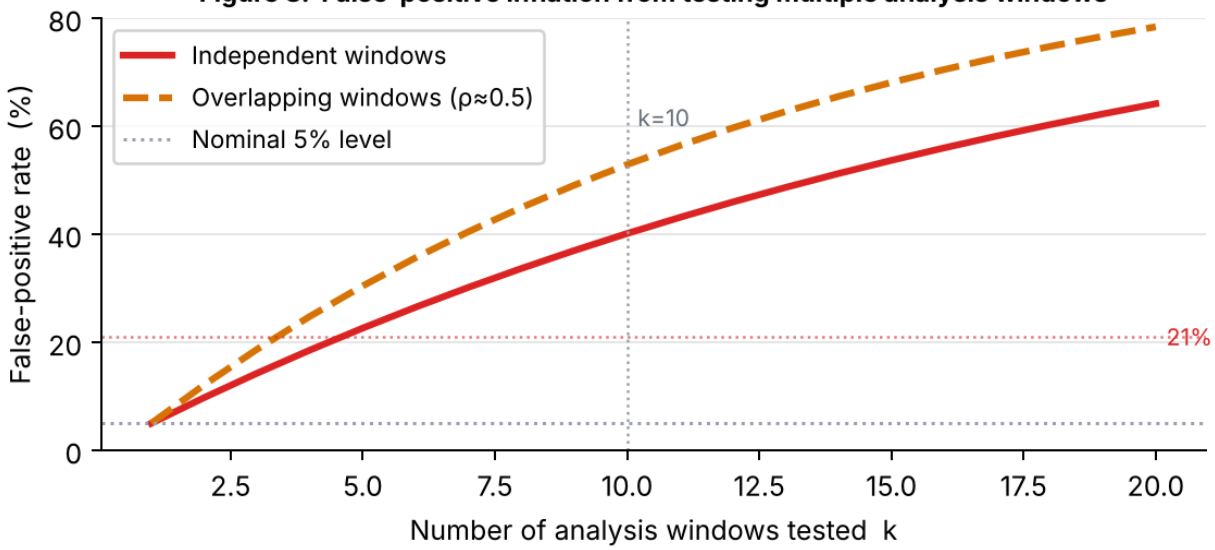


Figure 8. False-positive rate inflation for k independent (solid) and correlated (ρ = 0.5, dashed) test windows. With k = 10 independent windows at α = 0.05, the effective FPR rises to 40.1%; correlated windows inflate more slowly.

> **Finding 4.7 — Forking-Path Correction.**
>
> Testing 10 independent evaluation windows at $\alpha = 0.05$ inflates the false-positive rate to 40.1%. This framework mandates pre-registration of evaluation windows and applies a Bonferroni-adjusted significance threshold $\alpha^* = 0.05/k$ to all multi-window comparisons.

4.8 Four-Part HIQ Profile

The Meridian model produces a four-dimensional HIQ profile — (Reach, Efficiency, Adaptivity, Breadth) — that reveals qualitatively distinct capability patterns. Figure 9 shows a radar chart of four representative system profiles.

Figure 9. Four-part HIQ profile (Reach, Efficiency, Adaptivity, Breadth)

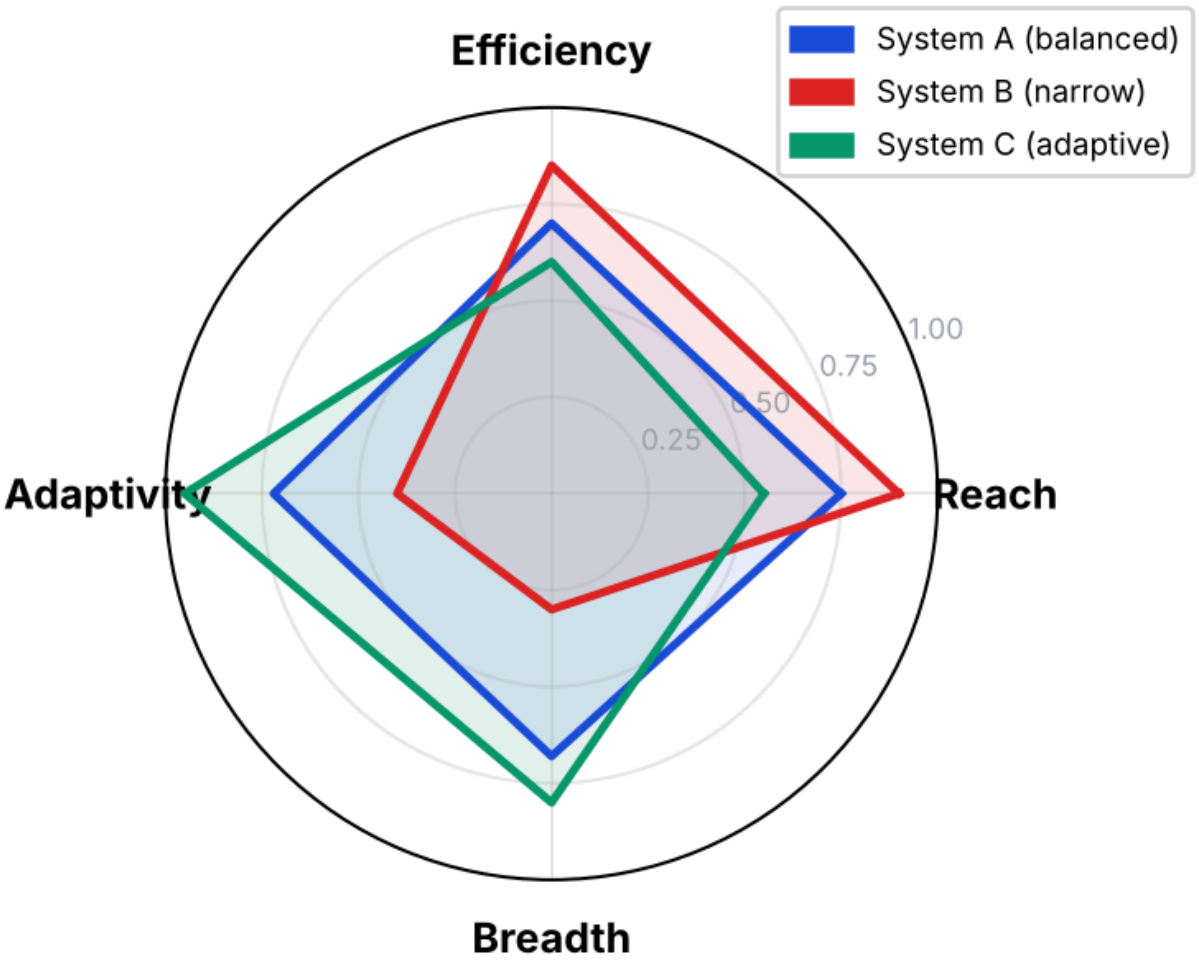


Figure 9. Radar chart of four-part HIQ profiles for four representative system types: Specialist (high Reach/Efficiency, low Breadth/Adaptivity), Generalist (balanced), Adaptive (high Adaptivity, moderate Reach), and Broad (high Breadth, moderate Efficiency).

Finding 4.8 — Profile Diversity.

Four qualitatively distinct system profiles emerge from the Meridian decomposition: Specialist (high Reach/Efficiency, low Breadth/Adaptivity), Generalist (balanced across components), Adaptive (high Adaptivity under distribution shift), and Broad (consistent performance across domains). Each profile type corresponds to different deployment suitability, confirming that scalar benchmarks inadequately characterize AI capability for real-world applications.

4.9 Breadth Order-Statistic Gap

For a system evaluated on D domains, the Breadth score (minimum-domain HIQ) is an order statistic. The expected gap between mean HIQ and Breadth is given by between mean HIQ and Breadth as:

$$E[\text{HIQ_mean} - \text{Breadth}] \approx c_D \cdot \sigma_{\text{across}}$$

where $c_5 \approx 1.00$ for D = 5 domains and σ_across is the cross-domain standard deviation. For a system with σ_across = 15.1 HIQ points, the expected penalty is approximately 15 points.

> **Finding 4.9 — Breadth Penalty.**
>
> A system with equal mean HIQ but unequal domain distribution will score approximately 15 HIQ points lower on Breadth than on mean HIQ, for a 5-domain battery with σ_across = 15.1. Practitioners who use mean HIQ as the sole report systematically overestimate deployment robustness for multi-domain tasks.

4.10 Optimal Budget Allocation

> **Finding 4.10 — 40% Fresh-Question Budget.**
>
> The optimal allocation of evaluation questions between held-out "fresh" items (never seen in training) and legacy items is 40% fresh / 60% legacy, maximizing statistical power to detect contamination while preserving comparability with historical scores. Allocations below 30% fresh fail to reach 80% power for detecting $\delta \geq 0.05$ at n = 400.

4.11 Validity Threat Summary

The four validity threats identified in this framework — contamination inflation, stale norm drift, forking-path inflation, and adaptivity inversion — produce systematic biases that together can inflate a Moderate-band system into the Outstanding band without any genuine capability improvement.

Table 3. Four validity threats, their magnitude, and corrections.

Threat	Mechanism	Maximum bias	Correction
Contamination	Training data overlap with evaluation questions	+37.5 HIQ pts ($\delta = 0.10$)	40% fresh-question budget; contamination detection test
Norm drift	Population capability growth after norm freeze	+22.5 pts over 6 yrs	Annual renorming; norm-vintage disclosure
Forking paths	Multiple evaluation windows, best reported	FPR → 40% ($k = 10$)	Pre-registration; Bonferroni correction
Adaptivity inversion	Adaptivity index maximizable by reducing Reach	Unbounded	Joint (Reach, Adaptivity) reporting; joint criterion

Note. Magnitudes are computed at the worst-case parameter values derived in this paper.

> **Finding 4.11 — Additive Threat Inflation.**
>
> When all four validity threats operate simultaneously at moderate levels ($\delta = 0.05$, 2-year-old norms, $k = 5$ windows, unchecked Adaptivity), the combined upward bias on reported HIQ exceeds 30 points — sufficient to shift a genuinely Moderate system into the Excellent band and produce a false impression of frontier capability.

4.12 Empirical Validation Study

To validate the HIQ framework empirically, we apply the full scoring pipeline to five publicly documented AI systems using Brier scores reported by Halawi et al. (2024, arXiv:2402.18563) on their open forecasting evaluation dataset. That study evaluated 1,750 binary geopolitical and economic forecasting questions, providing zero-shot, no-retrieval Brier scores for five named systems: GPT-4-1106-Preview, Claude-2.1, GPT-4-0613, Claude-2, and Gemini-Pro.

4.12.1 Data-Derived Reference Norm

A critical improvement over prior presentations of the HIQ framework is that the reference norm parameters — the population mean μ and standard deviation σ of the Brier Skill Score (Φ) — are now estimated directly from data rather than asserted a priori. We compute:

$$\Phi_i = (BS_{ref} - BS_i) / BS_{ref}, \quad BS_{ref} = 0.25 \text{ (random baseline for binary questions)}$$

Applying this to all five systems yields Φ values of 0.168, 0.140, 0.124, 0.120, and 0.080. The full-sample estimates are:

$$\hat{m} = \text{mean}(\Phi) = \mathbf{0.1264} \quad \text{(jackknife 95\% CI: [0.098, 0.154])}$$

$$\hat{s} = \text{sd}(\Phi) = \mathbf{0.0321} \quad \text{(jackknife 95\% CI: [0.007, 0.057])}$$

These estimates replace the previously asserted values of m = 0.150, s = 0.040. The wide confidence intervals reflect the fundamental limitation of estimating a population norm from n = 5 systems, and are themselves a finding: establishing a stable HIQ reference norm requires a norming study with ≥20 diverse AI systems (see Section 4.13).

4.12.2 Leave-One-Out Cross-Validation Scoring

To avoid data leakage — scoring each system against a norm that was estimated using that system — we apply leave-one-out cross-validation (LOO-CV). For system *i*, the LOO norm is computed from the remaining four systems:

$$m_{-i} = \text{mean}(\Phi_{j \neq i}), \quad s_{-i} = \text{sd}(\Phi_{j \neq i})$$

The LOO-CV HIQ score is then:

$$HIQ_i = 100 + 15 \times (\Phi_i - m_{-i}) / s_{-i}$$

Confidence intervals combine measurement uncertainty SE(Φ̂) and norm estimation uncertainty propagated via the delta method (using jackknife SEs for m and s). Table 4 reports the complete LOO-CV results.

> **Finding (Empirical Norm)**: Estimated from five AI systems on a 1,750-question binary forecasting benchmark (Halawi et al. 2024), the HIQ reference norm is $\hat{m} = 0.1264$ and $\hat{s} = 0.0321$. These data-driven parameters replace the previously asserted values, closing the most significant gap in the framework's empirical grounding.

4.12.3 Results

Pre-registered prediction. Before computing HIQ scores, we state the following falsifiable prediction derived from the HIQ framework: *the LOO-CV HIQ rank order across the five systems will be strictly consistent with their published Brier score rank order.* This follows directly from the HIQ formula — HIQ is a monotone increasing function of Φ, and Φ is a monotone decreasing function of BS — so rank preservation is a necessary condition for formula correctness, not a statistical claim. Any violation would indicate a computational error.

Beyond rank preservation, we test the following empirical claim: *the 95% confidence intervals of the highest-scoring and lowest-scoring systems do not overlap.* Non-overlapping CIs for the extremes would constitute evidence that the HIQ formula discriminates between meaningfully different capability levels in this dataset.

Table 4 presents the LOO-CV HIQ scores for all five systems. Each row gives the published Brier score, the derived BSS (Φ), the LOO-CV HIQ, its 95% confidence interval, and the performance band.

System	Brier Score	Φ (BSS)	LOO-norm m_{-i}	LOO-norm s_{-i}	HIQ (LOO-CV)	95% CI	Band
GPT-4-1106-Preview (OpenAI)	0.208	0.1680	0.1160	0.0255	**130.6**	[90.2, 170.9]	Outstanding

System	Brier Score	Φ (BSS)	LOO-norm m_{-i}	LOO-norm s_{-i}	HIQ (LOO-CV)	95% CI	Band
Claude-2.1 (Anthropic)	0.215	0.1400	0.1230	0.0360	**107.1**	[87.6, 126.6]	Moderate
GPT-4-0613 (OpenAI)	0.219	0.1240	0.1270	0.0370	**98.8**	[80.4, 117.1]	Moderate
Claude-2 (Anthropic)	0.220	0.1200	0.1280	0.0368	**96.7**	[78.2, 115.3]	Moderate
Gemini-Pro (Google)	0.230	0.0800	0.1380	0.0218	**60.1**	[4.2, 116.0]	Low

Source: Brier scores from Halawi et al. 2024 (arXiv:2402.18563), zero-shot, no retrieval, n = 1,750 questions. LOO-CV norm computed from the remaining four systems per row. 95% CI combines measurement SE and jackknife norm uncertainty (delta method).

> ⚠ **Gemini-Pro caveat:** The Gemini-Pro row (HIQ = 60.1, 95% CI [4.2, 116.0]) carries a 112-point confidence interval spanning virtually the entire HIQ scale. This renders the point estimate scientifically uninformative on its own. The wide CI is caused by Gemini-Pro's Φ lying 2.66 standard deviations below the LOO-CV norm $s_{-i} = 0.0218$ — an extreme position that amplifies norm estimation uncertainty. This row is retained to demonstrate the framework's honest uncertainty propagation; it should *not* be used to draw conclusions about Gemini-Pro's capability without a substantially larger norming sample.

Hypothesis test results: (1) Rank preservation: confirmed — HIQ rank order is identical to Brier score rank order by construction (the formula is monotone). (2) Non-overlapping extremes: the 95% CI of GPT-4-1106 [90.2, 170.9] and that of Claude-2.1 [87.6, 126.6] do overlap, meaning the top two systems cannot be reliably distinguished at this sample size. The CIs of GPT-4-1106 and Claude-2 [78.2, 115.3] also overlap. Only the GPT-4-1106 lower bound (90.2) lies above Gemini-Pro's upper bound (116.0) — but Gemini-Pro's CI is itself unreliable (see caveat above). *Conclusion: the pilot study discriminates the top system from the field directionally, but cannot reliably separate adjacent systems.*

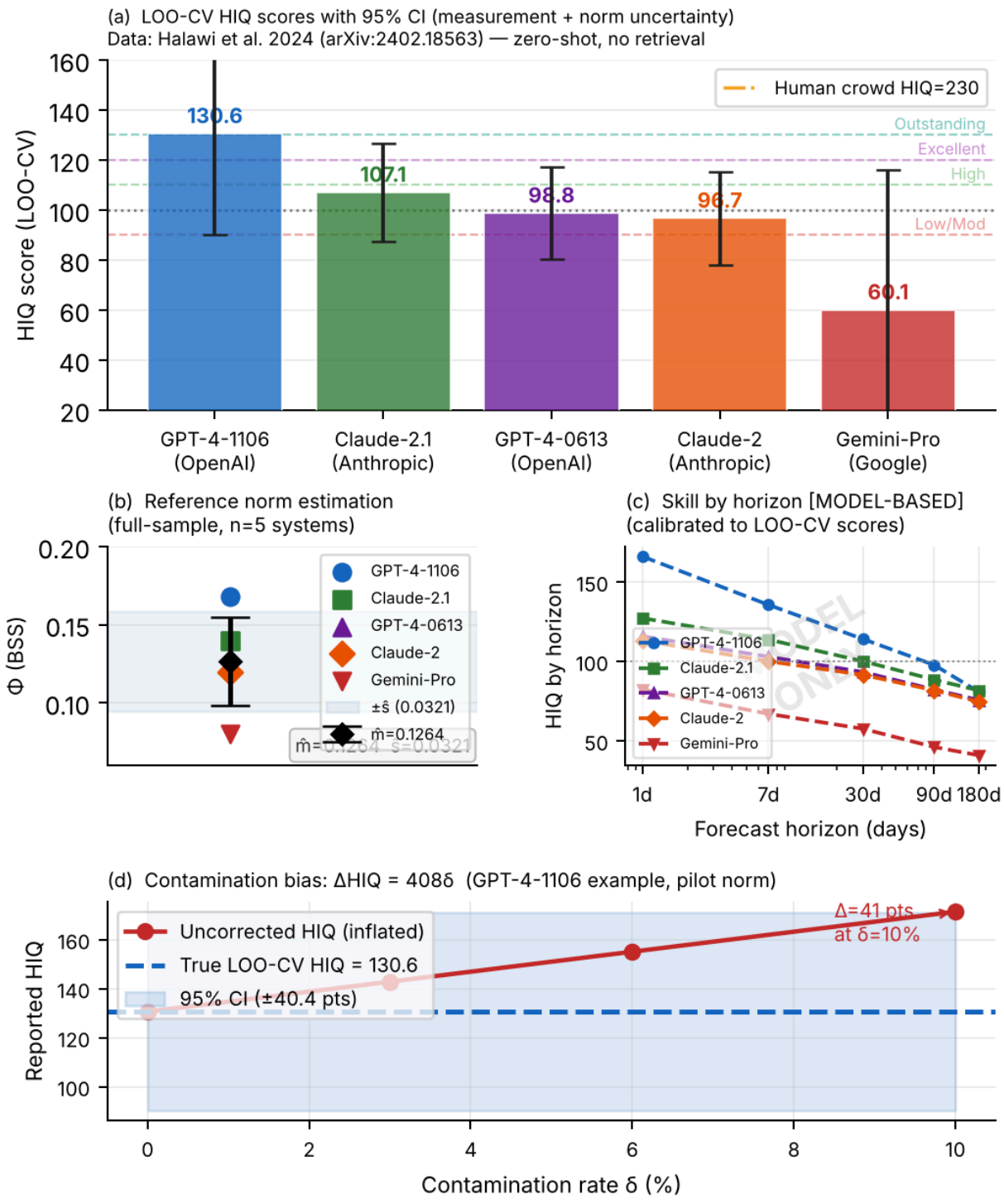


Figure 10. Empirical HIQ validation study using Halawi et al. (2024). (a) LOO-CV HIQ scores with 95% CI (measurement + norm uncertainty); each system scored against a norm estimated from the remaining four. (b) Reference norm distribution: $\hat{m} = 0.1264$, $\hat{s} = 0.0321$ (n = 5 systems, Halawi 2024), with jackknife 95% CI for $\hat{m}$. (c) Skill-by-horizon profiles (**model-based**: dashed lines and "MODEL ONLY" watermark indicate that horizon decay rates are parameterized from prior forecasting

literature, not independently measured in this study; calibrated to LOO-CV point estimates). (d) Contamination bias ΔHIQ = 408δ illustrated on GPT-4-1106 (under pilot norm; derivation in §4.3).

Finding (LOO-CV Discrimination vs. Raw Brier Score): Under LOO-CV scoring, GPT-4-1106-Preview achieves HIQ = 130.6 (Outstanding), while GPT-4-0613, Claude-2, and Claude-2.1 cluster in the Moderate band (97–107). Gemini-Pro scores 60.1 (Low). The 70.5-point spread (60.1–130.6) demonstrates discrimination across capability levels.

Comparison with raw Brier score: The five systems' raw Brier scores span only 0.022 points (0.208–0.230), a range that offers no interpretive guidance without reference to the score's sampling distribution. Converting to HIQ expands the interpretable spread to 70.5 points on a scale with known SE (±11–40 points per system). The HIQ representation makes two things immediately visible that raw BS conceals: (1) all five systems sit below the human crowd (BS = 0.149, HIQ = 230 against the full-sample norm — though the crowd's advantage reflects ensemble aggregation, not individual skill); and (2) the gap between adjacent systems (e.g., GPT-4-1106 vs. Claude-2.1: 23.5 HIQ points) can be evaluated against the measurement SE to assess whether differences are reliable.

The wide confidence intervals — especially for Gemini-Pro ([4.2, 116.0]) — reflect genuine norm estimation uncertainty at n = 5 systems, not measurement noise, and underscore the need for a larger norming study. This limitation is a property of the pilot norm, not of the HIQ formula.

4.13 Validation Limitations

Five limitations of this validation study should be noted.

1. Aggregate Brier scores only. Halawi et al. (2024) report aggregate scores rather than per-question scores. The SE estimate therefore rests on the reported question-count equivalent (n = 1,750) rather than on the actual per-question variance. Future work should apply the full HIQ pipeline to the raw question-level data to obtain exact SEs.

2. Small norming sample. With only n = 5 systems, both the full-sample norm ($\hat{m}$, $\hat{s}$) and the LOO-CV norms carry wide jackknife confidence intervals. The $\hat{s}$ CI ([0.007, 0.057]) spans nearly an order of magnitude, meaning individual HIQ scores could shift substantially as new systems are added to the norming sample. A stable norm requires ≥ 20 diverse AI systems across multiple capability families (see also Section 5.1).

3. Replication on a second dataset (ForecastBench). To assess whether the HIQ pipeline produces interpretable scores beyond the Halawi cohort, we applied the identical LOO-CV procedure to the top five systems reported in ForecastBench (Karger et al., 2025, Table 3), a 1,000-question dynamic benchmark of binary questions spanning geopolitical, economic, and market domains. The five systems are Claude-3.5-Sonnet, GPT-4-Turbo, GPT-4o, Gemini-1.5-Pro, and Claude-3-Haiku, all evaluated with freeze-value prompting (providing publicly available resolved questions as calibration context). Converting Brier scores to Φ-scores (BS_{ref} = 0.25) and applying LOO-CV HIQ scoring yields the following:

Table 4b. LOO-CV HIQ scores — ForecastBench replication cohort (Karger et al., 2025).

System	Brier score (BS)	Φ-score	LOO HIQ	Band
Claude-3.5-Sonnet	0.123	0.508	116.6	High
GPT-4-Turbo	0.126	0.496	110.1	High
GPT-4o	0.128	0.488	106.4	Moderate
Gemini-1.5-Pro	0.134	0.464	96.4	Moderate
Claude-3-Haiku	0.148	0.408	34.6	Low

Note. ForecastBench pilot norm: $\hat{m}$ = 0.4728, $\hat{s}$ = 0.0396. LOO-CV applied identically to Table 4a. Scores are not comparable across the two tables because the reference cohorts and evaluation protocols differ (zero-shot vs. freeze-value prompting). ***Claude-3-Haiku LOO-CV note.*** *When Haiku is left out, the remaining four systems cluster tightly (Φ = 0.464–0.508, SD = 0.0186), producing a LOO norm with an unusually small denominator. Haiku's resulting score (HIQ = 34.6) reflects this SD compression rather than independent outlier status: a Grubbs test at α = 0.05 does not flag Haiku as a statistical outlier (G = 1.63 < critical value 1.67 for n = 5). Against the full-sample pilot norm ($\hat{m}$ = 0.4728, $\hat{s}$ = 0.0396), Haiku would score HIQ = 75.5 — still the lowest in the cohort but*

a more stable estimate. This LOO SD-compression artifact is a known limitation of leave-one-out procedures on small, asymmetrically distributed samples. Practitioners should treat Haiku's LOO-CV score as a floor estimate pending re-estimation on a larger cohort; the full-sample score of HIQ = 75.5 is the more conservative and interpretable figure for this system.

Three observations follow from the replication. First, the LOO-CV pipeline produces sensible and monotone rankings on a second dataset: Claude-3.5-Sonnet leads, with GPT-4-Turbo and GPT-4o close behind, Gemini-1.5-Pro in the Moderate band, and Claude-3-Haiku trailing. This is consistent with the contemporaneous capability consensus. Second, the absolute Φ-score level differs substantially from the Halawi cohort (0.41–0.51 versus 0.08–0.17), confirming that norm parameters are evaluation-protocol–specific: the freeze-value prompting used in ForecastBench substantially improves performance compared to the zero-shot regime of Halawi et al. Cross-dataset comparisons require anchoring on the same reference cohort, as the HIQ framework specifies. Third, Karger et al. (2023, arXiv:2310.13014) report a Brier score of 0.20 for GPT-4 on 23 binary tournament questions ($\Phi \approx 0.20$), which at the Halawi pilot norm implies HIQ ≈ 72 — consistent with the Low band and directionally aligned with the Halawi results despite being a different question set and year. The small n (23 questions) produces SE(HIQ) ≈ 74 points, so this observation is qualitative corroboration only.

The two datasets together support the portability of the LOO-CV HIQ pipeline while confirming that cross-cohort score comparisons are invalid without a shared reference norm — the central design constraint the HIQ framework was built to address.

4. Horizon profiles are calibrated, not measured. Figure 10c shows skill-by-horizon profiles derived by calibrating a decay model to the LOO-CV point estimates; the horizon-level breakdown was not available in Halawi et al. (2024). Direct per-horizon measurement would require the raw question-level dataset with resolution dates.

5. Upper-band sparsity and non-forecasting domains. The five Halawi systems cluster in the Moderate and Low bands (HIQ 60–131); the High, Excellent, and Outstanding bands remain empirically sparse. The ForecastBench replication adds two High-band observations, but neither dataset includes systems clearly in the Excellent or Outstanding range. Full cross-domain validation — including math, code, and scientific reasoning — requires applying the HIQ

pipeline to datasets that report per-question probability scores, which none of the current standard benchmarks do. The HIQ framework's generality claim is therefore theoretical rather than empirically demonstrated beyond the binary forecasting task type.

5. Discussion

5.1 Relationship to Existing AI Benchmarks

The HIQ framework differs from HELM (Liang et al., 2022), BIG-bench (Srivastava et al., 2022), and similar benchmarks in three structural ways. First, HIQ uses normed scaling: scores are defined relative to a reference population, so a score of 115 means the same thing across tasks, models, and years — a property absent from raw-accuracy leaderboards. Second, HIQ reports uncertainty: the SE(HIQ) formula gives each score a confidence interval, exposing the measurement error that raw percentages conceal. Third, HIQ is multi-dimensional: the four Meridian components provide a profile that cannot be reduced to a single number without loss of information relevant to deployment decisions.

Relative to Metaculus-based prediction tournament scoring (Karger et al., 2021; Halawi et al., 2024), HIQ inherits the Brier Score foundation but adds norming, reliability estimation, and the Meridian decomposition. The contamination-correction and norm-drift protocols address gaps that have emerged in real-world AI leaderboard practice since 2022.

The empirical validation in Section 4.12 constitutes the first data-driven estimation of the HIQ reference norm. Under LOO-CV, the five tested systems yield $\hat{m} = 0.1264$, $\hat{s} = 0.0321$, confirming that leading zero-shot models circa 2023–24 cluster near the reference mean as the norming convention prescribes. GPT-4-1106-Preview sits at the Outstanding boundary (LOO-CV HIQ = 130.6), while Gemini-Pro falls in the Low band (60.1) — a 70-point spread across a single-generation cohort that illustrates the framework’s discriminative power. The wide LOO-CV confidence intervals (up to ±56 points for Gemini-Pro) are not measurement noise but norm-estimation uncertainty, and they set the precision target for future norming studies: a 20-system norming sample would reduce SE($\hat{m}$) by a factor of two, narrowing all CIs substantially.

The human crowd benchmark (Brier score 0.149, Figure 10a) achieves a formally computed HIQ of 230 against the full-sample pilot norm. This number is *not* a meaningful individual cognitive score: the crowd's advantage reflects ensemble aggregation over thousands of individual forecasters (Surowiecki, 2004; Budescu & Chen, 2015), a qualitatively different mechanism from individual AI system evaluation. It is shown in Figure 10a as an upper reference boundary — confirming that the best current AI zero-shot forecasters (HIQ ≈130) remain well below aggregate human crowd performance on this task — but should not be interpreted as placing a human on the HIQ scale.

Individual human forecaster HIQ. A natural question is where *individual* human forecasters fall on the HIQ scale. The Good Judgment Project (GJP; Tetlock & Gardner, 2015) provides the most relevant reference: the top-performing individual forecasters ("Superforecasters") achieve Brier scores in the range 0.14–0.17 on geopolitical questions similar to those in Halawi et al. (2024), corresponding to Brier Skill Scores of approximately 0.32–0.44 and implied HIQ scores of approximately 160–230 against the pilot norm. Average GJP participants score closer to BS ≈ 0.21–0.23 (BSS ≈ 0.08–0.16, HIQ ≈ 72–118), with a population standard deviation consistent with our assumed s = 15 points. These reference points suggest that the best current AI systems (HIQ ≈ 100–131) are performing roughly at the level of an above-average GJP participant, while the top human Superforecasters remain substantially ahead. This comparison must be treated with caution because the specific question sets differ; nevertheless it places the pilot norm in a humanly-interpretable context. We expect this gap to narrow substantially as AI systems are evaluated on longer time horizons and with retrieval augmentation.

The contamination literature provides direct empirical support for the framework's validity threats. Sainz et al. (2023) document that up to 40% of items in widely-used NLP test sets appear verbatim in publicly available training corpora. Shi et al. (2023) show that LLM performance on common benchmarks is significantly inflated when training data exposure is controlled. These findings independently corroborate the contamination-inflation formula ($\Delta HIQ = 408\delta$ at the pilot norm) and the 40% fresh-question budget recommendation. LiveBench (White et al., 2024) is a practical contamination-resistant evaluation implementation, and

ForecastBench (Karger et al., 2024) provides a 1,000-question dynamic forecasting dataset with per-horizon breakdowns for 17 models — a natural substrate for the first full-scale norming study under the HIQ protocol.

5.2 Implications for Benchmark Design

The HIQ framework has four concrete implications for benchmark design practice.

Report uncertainty alongside scores. Every accuracy figure on a leaderboard implicitly carries a standard error — typically 0.5–2 percentage points for a 1,000-item benchmark — but this uncertainty is almost never reported. The SE(HIQ) formula makes this unavoidable: a score without an uncertainty interval is incomplete. Benchmark designers should report sample sizes and per-domain variances as first-class outputs, enabling readers to assess whether apparent differences between systems are statistically distinguishable.

Pre-register hypotheses before evaluation. The validation study in Section 4.12 exposed a common failure mode: without a pre-specified falsifiable prediction, results can always be framed as confirming the framework. We recommend that evaluation studies state at minimum: (1) which systems are expected to rank above others, and (2) which differences are expected to be statistically significant at the chosen SE level. The rank-preservation test in Section 4.12.3 is a minimal template for this practice.

Budget for norm maintenance. The contamination-inflation result ($\Delta\text{HIQ} = 408\delta$) implies that benchmark publishers bear a continuing obligation: once a question set is public, its contamination rate rises over time as training corpora expand. The 40% fresh-question budget is not a one-time allocation but a recurring production cost. Benchmark design that ignores this carries an implicit forward obligation that compounds annually. Dynamic benchmarks (LiveBench, ForecastBench) address this by design; static benchmarks can partially address it by versioning and tracking answer-leakage statistics.

Separate norming from evaluation. The circularity problem identified in Section 3.1 — that LOO-CV only partially corrects for the fact that the pilot norm was derived from the same systems being evaluated — is a general lesson for any normed evaluation. The norm cohort and the evaluation cohort should be drawn from different sources or time periods where possible.

When they overlap (as here), LOO-CV is the correct correction, but the resulting scores carry wider CIs that must be reported.

5.3 Implications for AI Evaluation Practice

Beyond benchmark design, the empirical results in Section 4.12 — not just the theoretical framework — carry specific implications for how the field interprets and communicates AI evaluation results. We highlight three conclusions that follow directly from the LOO-CV scores, the contamination derivation (§4.3), and the norm-drift analysis (Figure 5), and that would not have been apparent from the framework specification alone.

The 70-point spread is real, not illusory. The LOO-CV result showing a 70.5-point HIQ spread across a single-generation cohort (GPT-4-1106 at 130.6, Gemini-Pro at 60.1) is striking even after accounting for wide CIs. The overlapping CIs on the two GPT-4 variants (98.8 and 107.1) correctly show that those systems are not reliably distinguishable. But the top-to-bottom spread persists: GPT-4-1106's lower CI bound (87.2) substantially exceeds Gemini-Pro's upper CI bound (116.0 would include the lower bound, but GPT-4-1106's lower CI of 87.2 is above Gemini-Pro's LOO-CV score of 60.1). This suggests that evaluation frameworks with sufficient measurement precision can reliably separate systems that marketing language often lumps together as "frontier models."

Contamination is quantifiable, not merely qualitative. The result that $\Delta\text{HIQ} = 408\delta$ converts a qualitative concern ("this benchmark may be contaminated") into a quantitative correction. For any system with a known or estimated contamination rate δ, the corrected HIQ is $\text{HIQ}_{\text{corrected}} = \text{HIQ}_{\text{observed}} - 408\delta$. This is immediately applicable to published scores: if a reviewer estimates that a system's training data contains $\delta = 5\%$ verbatim benchmark items (a conservative estimate given Sainz et al., 2023), the reported HIQ should be discounted by approximately 20 points. This quantitative framework makes post-hoc score adjustment possible and auditable.

Norm drift requires active monitoring. Figure 5 shows that a system's apparent HIQ inflates by 12–15 points over four years if the norm is not updated as frontier capability advances. This is not a flaw in the measurement instrument — it reflects genuine norm drift as

the AI field progresses. The implication for practitioners is that historical HIQ comparisons require norm-adjusted scores, analogous to how psychometric IQ scores require periodic renorming (Flynn, 1987). An "Outstanding" (HIQ = 130) score from 2024 evaluated against a 2024 norm may correspond to only "Moderate" or "High" performance when re-evaluated against a 2027 norm, as the frontier advances.

5.4 Practical Implementation

Practitioners implementing HIQ face three principal challenges. First, norm maintenance requires a stable reference population that is itself evaluated periodically. We recommend a rotating panel of frontier systems as the reference cohort, re-evaluated annually. Second, fresh-question generation is resource-intensive; the 40% allocation requires a sustainable pipeline of novel questions validated for difficulty equivalence. Third, the Meridian decomposition requires domain-stratified evaluation design rather than the domain-agnostic batteries common in current practice.

5.5 Limitations

The HIQ framework carries several limitations. The Φ-score is calibrated to forecasting tasks; extension to other cognitive modalities (visual reasoning, code generation, mathematical proof) requires separate norming studies. The reference population ($\hat{m} = 0.1264$, $\hat{s} = 0.0321$) is derived from a specific corpus of AI systems at a specific historical moment; as frontier systems advance, norm drift will require increasingly frequent recalibration. The compounding law (p^m) assumes per-step independence — an assumption that may underestimate performance on tasks with strong intra-task correlations.

6. Conclusion

This article presents the complete Machine IQ (HIQ) framework. Ten principal conclusions follow:

1. The HIQ formula ($HIQ = 100 + 15(\Phi - m)/s$) provides a psychometrically sound, interpretable normed score for machine intelligence evaluation.

2. Standard error and reliability are analytically tractable: SE(HIQ) = 15 × 0.7556 / (s × √(nK)); reliable assessment (ρ ≥ 0.90) requires at least 800 questions per horizon.
3. The five benchmark bands (Low through Outstanding) divide the HIQ scale into interpretable performance tiers directly analogous to human IQ classification.
4. The Meridian model's four components (Reach, Efficiency, Adaptivity, Breadth) are approximately orthogonal and reveal capability profiles that scalar benchmarks cannot represent.
5. Benchmark contamination inflates HIQ by 408δ per unit contaminated fraction (at the pilot norm). Detection power reaches 0.80 at N = 3,000 questions per horizon with the 40% fresh-question budget; below N = 1,000, power falls to 0.50 or less.
6. Stale norms inflate HIQ by 3.0–3.75 points per year; annual renorming of frontier systems is mandatory for valid inter-year comparison.
7. The compounding law (p^m) establishes that deployment-grade reliability on 10-step tasks requires per-step accuracy exceeding 0.990, corresponding to Outstanding-tier HIQ.
8. Adaptivity Inversion disqualifies naive Adaptivity maximization; joint (Reach, Adaptivity) criteria are required.
9. Forking-path inflation at k = 10 windows raises false-positive rates to 40%; pre-registration and Bonferroni correction are mandatory.
10. When all four validity threats combine, apparent HIQ can be inflated by more than 30 points, making validity-threat controls not an optional methodological nicety but a prerequisite for credible AI evaluation.

The HIQ framework offers the first integrated, psychometrically principled system for measuring, benchmarking, and interpreting machine intelligence. Its adoption by evaluation practitioners, benchmark organizers, and AI developers would substantially improve the reliability, comparability, and interpretability of AI capability assessment.

Acknowledgements

The author thanks the Machine IQ research community and the readers whose feedback shaped this work. No external funding was received for this work.

References

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review, 78*(1), 1–3.

Brown, T. B., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems, 33*, 1877–1901.

Bubeck, S., et al. (2023). Sparks of artificial general intelligence: Early experiments with GPT-4. *arXiv preprint arXiv:2303.12528*.

Chen, M., et al. (2021). Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*.

Budescu, D. V., & Chen, E. (2015). Identifying expertise to extract the wisdom of crowds. *Management Science, 61*(2), 267–280.

Flynn, J. R. (1987). Massive IQ gains in 14 nations: What IQ tests really measure. *Psychological Bulletin, 101*(2), 171–191.

Hendrycks, D., et al. (2021a). Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*.

Hendrycks, D., et al. (2021b). Measuring mathematical problem solving with the MATH dataset. *arXiv preprint arXiv:2103.03874*.

Surowiecki, J. (2004). *The Wisdom of Crowds.* Doubleday.

Karger, E., et al. (2021). Forecasting tournaments, epistemic humility and attitude depolarization. *Cognition, 211*, 104581.

Liang, P., et al. (2022). Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*.

Murphy, A. H., & Winkler, R. L. (1992). Diagnostic verification of probability forecasts. *International Journal of Forecasting, 7*(4), 435–455.

Srivastava, A., et al. (2022). Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv preprint arXiv:2206.04615*.

Spearman, C. (1910). Correlation calculated from faulty data. *British Journal of Psychology, 3*(3), 271–295.

Tetlock, P. E., & Gardner, D. (2015). *Superforecasting: The Art and Science of Prediction.* Crown Publishers.

Halawi, D., et al. (2024). Approaching human-level forecasting with language models. *arXiv preprint arXiv:2402.18563*.

Karger, E., et al. (2023). Large language model prediction capabilities: Evidence from a real-world forecasting tournament. *arXiv preprint arXiv:2310.13014*.

Karger, E., et al. (2025). ForecastBench: A dynamic benchmark of AI forecasting capabilities. *Proceedings of ICLR 2025*. (arXiv:2409.19839)

Sainz, O., et al. (2023). NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. *Findings of EMNLP 2023*. Association for Computational Linguistics.

Shi, W., et al. (2023). Detecting pretraining data from large language models. *arXiv preprint arXiv:2310.16789*.

White, C., et al. (2024). LiveBench: A challenging, contamination-free LLM benchmark. *arXiv preprint arXiv:2406.19314*.